

Predictive Modeling and Causal Inference for Prognostic Stratification in the Rotterdam Breast Cancer Cohort

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Abstract

Background: Accurate risk stratification is paramount in the clinical management of breast cancer, as long-term prognosis significantly influences treatment decisions. This study aims to identify key predictors of tumor recurrence and mortality and to evaluate the causal impact of adjuvant hormonal therapy within an observational framework.

Methods: Leveraging the Rotterdam Breast Cancer dataset ($n=2,982$), we implemented a multi-stage analytical pipeline. Predictive modeling was conducted using Logistic Regression with AIC-based feature selection and Stratified 5-Fold Cross-Validation. Survival analysis was performed using Kaplan-Meier estimators and Cox Proportional Hazards models. To address non-random treatment assignment, Inverse Probability of Treatment Weighting (IPTW) was employed to estimate the Average Treatment Effect (ATE) of hormonal therapy.

Results: Pathological features, specifically log-transformed nodal involvement and tumor size, emerged as the most significant predictors of adverse outcomes. The survival analysis highlighted violations of the proportional hazards assumption for specific covariates, necessitating stratified approaches. Causal inference through IPTW revealed a significant protective effect of hormonal therapy on recurrence-free survival once baseline clinical imbalances were adjusted.

Conclusions: The integration of predictive and causal frameworks allows for a robust separation of baseline risk from therapeutic effect. These findings underscore the importance of accounting for selection bias in real-world data to refine clinical guidelines and improve personalized patient management.